

UNIVERSITY OF THE WEST INDIES

Department of Computing

COMP3161 – Database Management Systems (Semester II, 2026)

Problem Definition – Part 1

You are required to create a course content management software for an educational institution. Below are the main features of the system.

- Admins should be able to create accounts for new users.
- Users should be able to login.
- Accounts can be of type Lecturer/Course Maintainer, Student, Admin
- Admins should be able to create a Course.
- A student can be assigned to courses.
- A Course should consist of Members(Students. Lecturer/Course Maintainer)
- A Course should consist of several Discussion Forums.
- A Discussion Forum should consist of several discussion threads.
- Students and Course Maintainers should be able to create or reply to discussion threads.
- Students should be able to reply to replies thereby creating another thread
- A Course should have Calendar Events e.g Assignment Due Date.
- The Course Maintainer should have the ability to create, remove and update calendar events.
- The content of a course should consist of sections/topics.
- Each section should consist of several section items e.g links, lecture slides.
- Students should be able to submit assignments.
- Course Maintainers should be able to add, update or remove section items or create new sections.

Draw an E-R diagram for the above application. Make sure that you state any assumptions that you made and any information that could not be captured in the diagram. You are free to add any additional information that you feel may be useful for this application. However, you must state why you choose to do so.

Bonus points will be assigned for any useful entities proposed. **Convert the E-R diagram into a set of relations/tables.** (NB: Do not attempt to normalize your tables at this time.)

DUE DATE: Mon, February 9, 2026 @ 11:59 PM.

No late assignment will be accepted.

A submission container will be made available at the appropriate time.

The document (including the ER diagram) must be prepared using any relevant software.

You must keep a copy of the ER diagram as you will need it to do the next part of the project

